

Less:

Rec:

Conditional probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \text{when } P(B) > 0$$

Multiplication rule

$$\begin{aligned} P(A_1 \cap A_2 \cap \dots \cap A_n) &= P(A_1) P(A_2|A_1) P(A_3|A_1 \cap A_2) \dots \\ &= \prod_{i=1}^n P(A_i | \bigcap_{j=1}^{i-1} A_j) \end{aligned}$$

Today: Total Probability theorem

Bayes rule

Independence, Conditional Independence.

$$P(A_1 \cap A_2 | C) = P(A_1 | C) P(A_2 | C \cap A_1)$$

LHS $\frac{P(A_1 \cap A_2 \cap C)}{P(C)}$

RHS $\frac{\cancel{P(A_1 \cap C)}}{P(C)} \frac{P(A_2 \cap C \cap A_1)}{\cancel{P(C \cap A_1)}}$

$$P(A_1 \cap A_2 \dots A_n | C) = P(A_1 | C) P(A_2 | A_1 \cap C) \dots P(A_n | \bigcap_{i=1}^{n-1} A_i \cap C)$$

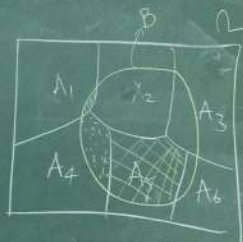
Total Probability theorem

Let $A_1, A_2, \dots, A_n$ that partition the sample space Ω
i.e., A_i, A_j are disjoint $\forall i \neq j$

$$\& \bigcup_{i=1}^n A_i = \Omega$$

Then for any event B

$$P(B) = \sum_{i=1}^n P(B \cap A_i)$$



$$\begin{aligned} B &= B \cap \Omega \\ &= B \cap \left(\bigcup_{i=1}^n A_i \right) = \bigcup_{i=1}^n (B \cap A_i) \end{aligned}$$

$$P(B) = P\left(\bigcup_{i=1}^n (B \cap A_i)\right)$$

additivity

$$= \sum_{i=1}^n P(B \cap A_i)$$

Suppose $A_1, A_2, \dots$ that partition Ω
 then also it follows that

$$\begin{aligned} A_0 &= [0, 1) \\ A_1 &= [-1, 0) \\ A_2 &= [1, 2) \end{aligned} \quad \Omega = \mathbb{R}$$

$$P(B) = \sum_{i=1}^{\infty} P(B \cap A_i)$$

$$A_n = \left\{ \underbrace{TT \dots T}_n H \right\} \quad \Omega = \{H, TH, \dots\}$$

Bayes rule

Example : A : aircraft is present
 B : event that alarm is signal.

$$P(B|A) = 0.99$$

$$P(B^c|A^c) = 0.9$$

$$P(A) = 0.05$$

} Design

Analogy

biomedical
testing

A : cancer

B : diagnosis indicates cancer

Find the probability that the aircraft was actually present given that alarm is signaled. $P(A|B)$

$$P(A^c) = 1 - P(A) = 0.95 \quad P(B|A^c) = 1 - 0.9 = 0.1$$

$$P(B^c|A) = 1 - P(B|A) = 0.01 \quad = 1 - P(B|A^c)$$

$$P(A^c|B) = 1 - P(A|B)$$

Proof: $P(A^c \cup A|B) = P(\Omega|B)$
 $= 1$

$$= P(A^c|B) + P(A|B)$$

$$\Rightarrow P(A^c|B) = 1 - P(A|B)$$

	B	B ^c	
A	B A	B ^c A	miss detection
A ^c	B A ^c ↓ false positive	B ^c A ^c	

Recap
 $P(A \cup A^c)$

$P(A)$

Bayes rule



Rem

$$P(A_1 \cup A_2 | B) = \frac{P((A_1 \cup A_2) \cap B)}{P(B)}$$

$$= \frac{P(A_1 \cap B)}{P(B)} + \frac{P(A_2 \cap B)}{P(B)}$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$= \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|A^c)P(A^c)}$$

$$P(B|A)P(A) + P(B|A^c)P(A^c)$$

Bayes
rule

$$P(A|B) = \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|A^c)P(A^c)}$$

$$P(A|B) = \frac{0.99 \times 0.05}{0.99 \times 0.05 + 0.1 \times 0.95}$$
$$= 0.34$$

In general if you have $A_1, A_2, \dots, A_n$ that partition Ω

$$\begin{aligned} P(A_i | B) &= \frac{P(A_i \cap B)}{P(B)} \\ &= \frac{P(A_i \cap B)}{\sum_{j=1}^n P(A_j \cap B)} \end{aligned}$$

$$P(A_i | B) = \frac{P(B | A_i) P(A_i)}{\sum_{j=1}^n P(B | A_j) P(A_j)} \quad \text{Bayes rule}$$

You can get paired opposite following Tennis players
Senia, Bopanna or Leander.

Your chances of beating them are 0.3, 0.4, 0.5
respectively. Chances of each of them getting
picked are 0.5, 0.25, 0.25 respectively.

Given that you won, what is the probability that
you won against Senia.

B : You win.

A_1 : Senia getting picked

$$P(A_1) = 0.5$$

A_2 : Bopanna "

$$P(A_2) = 0.25$$

A_3 : Leander "

$$P(A_3) = 0.25$$

$$P(B|A_3) = 0.5$$

$$P(B|A_1) = 0.3$$

$$P(B|A_2) = 0.4$$

$$P(A_1|B) = \frac{P(B|A_1)P(A_1)}{P(B|A_1)P(A_1) + P(B|A_2)P(A_2) + P(B|A_3)P(A_3)}$$

$$0.3 \times 0.5$$

$$0.3 \times 0.5 + 0.4 \times 0.25 + 0.5 \times 0.25$$

Independence

Events A and B are said to be independent if

$$P(A \cap B) = P(A) P(B)$$

$$\Leftrightarrow P(A|B) = P(A) \text{ if } P(B) > 0$$

$$\Leftrightarrow P(B|A) = P(B) \text{ if } P(A) > 0$$

$$\begin{aligned} P(A \cap B) &= P(A) P(B|A) \\ &= P(B) P(A|B) \end{aligned}$$

$$\Rightarrow P(A)P(A^c) - P(A)P(B) = 0$$

$$\Rightarrow P(A)[P(A^c) - P(B)] = 0$$

$$\Rightarrow P(A) = 0 \text{ or } P(A^c) = P(B)$$

Suppose you have A, B st $A \cap B = \emptyset$.

Can they be independent?

$$P(A \cap B) = 0 = P(A)P(B)$$

→ Not always true (ie, it is not true for any A, B)

True only if either $P(A) = 0$ or $P(B) = 0$

Exercise:

Suppose A, B are independent,
Are A, B^c independent?

Prove this!

Conditional Independence

A, B are said to be independent conditioned over C if

$$P(A \cap B | C) = P(A | C) P(B | C)$$

or equivalently

$$\Leftrightarrow P(A | B \cap C) = P(A | C) \text{ if } P(B \cap C) > 0$$

Conditional Independence

$\nRightarrow$ Independence

Independence $\nRightarrow$ Conditional Independence

A and B are independent conditioned over C

$\nRightarrow$ A, B are independent

Ex 1: Consider two independent coin tosses

$\{HH, HT\} = A$: first toss is head

$\{HH, TH\} = B$: second toss is head.

$\Omega = \{HH, HT, TH, TT\}$

$= \{H, T\} \times \{H, T\}$

$\{HT, TH\} = C$: event that the outcomes are different in the two tosses.

$$P(A \cap B) = P(\{HH\}) = \frac{1}{4} = P(A)P(B)$$

$$P(A) = \frac{2}{4} = P(B)$$

$\Rightarrow$ A, B are independent

(3)

Shows that
conditioning can increase or
decrease probability.

If $A \cap B \neq \emptyset$, $P(B) > 0$, $P(A) > 0$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \leq P(A)$$

If $A \subseteq B$

$$\begin{aligned} P(A|B) &= \frac{P(A \cap B)}{P(B)} \\ &= \frac{P(A)}{P(B)} \\ &\geq P(A) \end{aligned}$$

$$\begin{aligned} P(A|C) &= \frac{P(A \cap C)}{P(C)} \\ &= \frac{P(\{HT, TH\})}{P(\{HT, TH\})} \\ &= \frac{1}{2} = P(B|C) \end{aligned}$$

$$\begin{aligned} P(A \cap B|C) &= \frac{P(A \cap B \cap C)}{P(C)} \\ &= 0 \\ &\neq P(A|C)P(B|C) = \frac{1}{4} \end{aligned}$$



Ex2: In this experiment you start by choosing a coin among two options first that gives heads with probability 0.7 second that gives heads w.p. 0.2
pick any coin with probability 0.5

After picking the coin you toss it twice. Both the tosses are independent of each other

C: you pick first coin $\Rightarrow P(C) = 0.5$

A: first toss is head

B: second toss is head

$$\Omega = \{1, 2\} \times \{H, T\} \times \{H, T\} \\ = \{1HH, 1HT, \dots\}$$

$$A = \{1HH, 1HT, 2HH, 2HT\}$$

$$B =$$

$$P(A|C) = 0.7 = P(B|C)$$

C: you pick first coin $\Rightarrow P(C) = 0.5$

A: first toss is head

B: second toss is head

$$\Omega = \{1, 2\} \times \{H, T\} \times \{H, T\} \\ = \{1HH, 1HT, \dots\}$$

$$A = \{1HH, 1HT, 2HH, 2HT\}$$

$$B =$$

$$P(A|C) = 0.9 = P(B|C)$$

$$P(A|C^c) = P(B|C^c) \\ = 0.2$$

From the experiment
defn:

A is independent of
B conditioned over C

$$P(A) = P(A|C) + P(A|C^c) \\ = P(A|C)P(C) \\ + P(A|C^c)P(C^c)$$

$$P(A \cap B) = P(A \cap B|C) \\ + P(A \cap B|C^c)$$